

COURSE TITLE : ELECTRONICS AND INSTRUMENTS WORKSHOP
COURSE CODE : 4219
COURSE CATEGORY : A
PERIODS/WEEK : 6
PERIODS/SEMESTER: 84
CREDITS : 3

On completion of this course the student will be able to

1.1.0 Soldering Practice

1.1.1 solder and desolder electronic components on PCB.

1.2.0 Electric Wiring

1.2.1 one lamp controlled by one switch

1.2.2 Two lamp controlled by two switch (series)

1.2.3 Two lamp controlled by two switch(parallel)

1.2.4 stair case wiring

1.2.5 Hospital wiring

2.1.0 Instrument Wiring &Tagging

2.1.1 Practice Loop wiring & panel wiring

2.1.2 Wire instrument panel board with various accessories

2.1.3 Test assembled instrument loop wiring for various parameters and faults

2.1.4 Understanding Instrument Tagging ,Panel & Marshling Cab wiring, Junction Box Wiring, Cable Termination &Feruling Details

3.1.0 Relay Logic Practice

3.1.1 control a 230v/60w bulb using 12v relay

3.1.2 Start and stop of a 12v DC motor using relays(self hold method)

3.1.3 Run a 12v dc motor in Forward Direction & Reverse Direction using DPDT relay

4.1.0 Study of Instrument Fittings

4.1.1 Elbow,Tee,Union,conversion union,Reducing unon, Union Elbow,Union tee, connectors & nipples etc.

4.1.2 Threads – NPT,BSP,BSPT,BSPP etc

4.1.3 Study of Maniflods 2 way,3 way,5 way

4.1.4 Instrument hookups in Liquid ,Gas & Steam Lines

5.1.0 Tubing & Threading

5.1.1 Identify the tools for tubing & Threading works-Tube Cutter, Tube Bender Swage tool, Die, Tap etc

5.1.2 Perform Bending, flaring and glanding of metal tubes

6.1.0 Cable Glanding ,Shrouding

6.1.1 Identify Signal Cable,Control Cable and Power Cables

6.1.2 Identify the fittings for Cable Glanding & Perform Cable glanding &Shrouding

7.1.0 Troubleshooting:

7.1.1 Testing instrumentation loop, instrumentation panel, Perform loop checking

7.1.2 Familiarisation of hand held calibrators-HART and Foundation FIELDBUS

8.1.0 Servicing and Repairing of Instruments

8.1.1 Pressure gauges, transmitters, recorders, relays, regulators, control valves, pressure switches , level switches& control valve limit switches

9.1.0 Control valve

9.1.1 Dismantle & assemble valve to identify it's components as per sketch

10.1.0 Dismantle, assemble & calibrate pressure gauge .

11.1.0 Install any one instrument using screw type connection.

12.1.0 Install any one instrument using flange type connection

13.1.0 Install any one transmitter as per instrument hookup

14.1.0 Study, class of fire , extinguishing methods and first aids